

Design and Implementation of Programming Languages

You will learn programming language concepts by designing and implementing our own programming language system.

- We will define a programming language. For example, “factorial” is written in our language as follows:

```
let x = read in
letrec fact(n) =
  if iszero n then 1
  else ((fact (n-1)) * n)
in (fact x)
```

- We will design and implement an interpreter for the language:

Program $\rightarrow$ Interpreter $\rightarrow$ Result

- We will design and implement a type checker for the language:

Program $\rightarrow$ Type Checker $\rightarrow$ Safe/Unsafe